

Analyse et Prédiction des Maladies Cardiaques

Statistiques des modèles - Ensemble de test

Version corrigée - Avril 2025

Performances des modèles - Ensemble de test

Modèle	Accuracy	Precision	Recall	F1-Score	AUC	TP	TN	FP	FN
Logistic Regression	88.6%	82.2%	94.9%	88.1%	93.2%	37	41	8	2
Decision Tree	83.0%	80.0%	82.1%	81.0%	86.7%	32	41	8	7
KNN	88.6%	82.2%	94.9%	88.1%	91.9%	37	41	8	2
Random Forest	90.9%	86.0%	94.9%	90.2%	93.9%	37	43	6	2
AdaBoost	76.1%	70.5%	79.5%	74.7%	89.7%	31	36	13	8
SVM	90.9%	86.0%	94.9%	90.2%	92.0%	37	43	6	2
Voting Ensemble	92.0%	88.1%	94.9%	91.4%	93.4%	37	44	5	2

Détails - Logistic Regression

Métriques principales

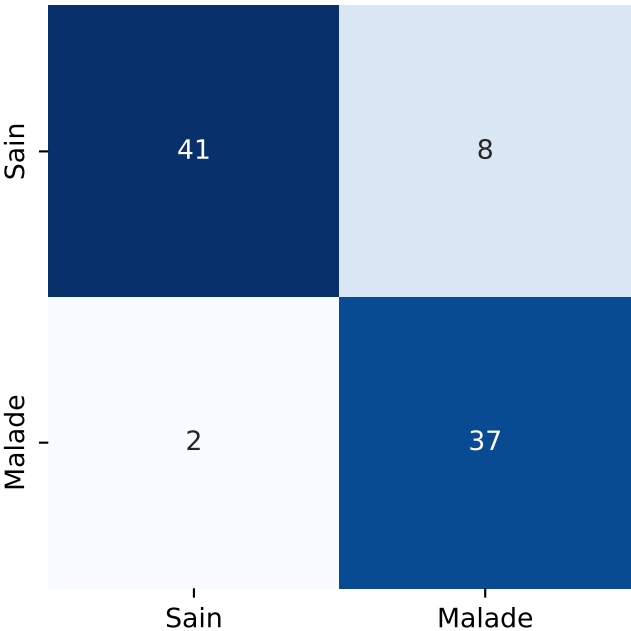
Accuracy: 88.6%

Precision: 82.2%

Recall: 94.9%

F1-Score: 88.1%

Matrice de confusion



Détail matrice

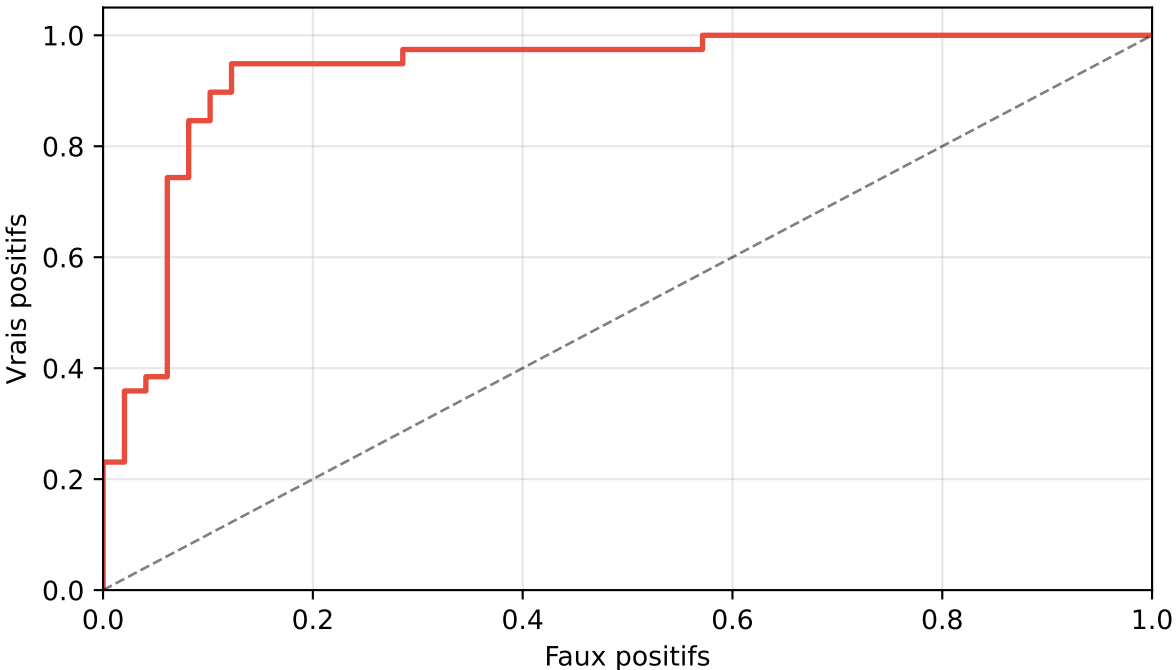
TP: 37

TN: 41

FP: 8

FN: 2

ROC (AUC = 0.932)



Détails - Decision Tree

Métriques principales

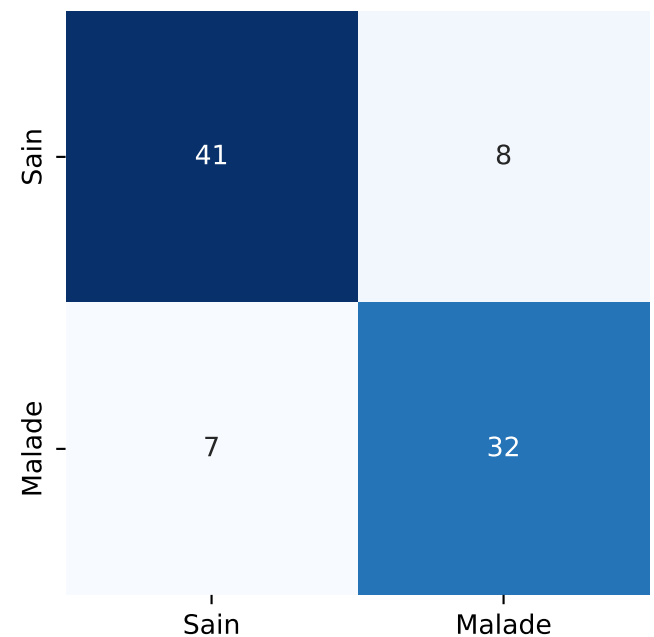
Accuracy: 83.0%

Precision: 80.0%

Recall: 82.1%

F1-Score: 81.0%

Matrice de confusion



Détail matrice

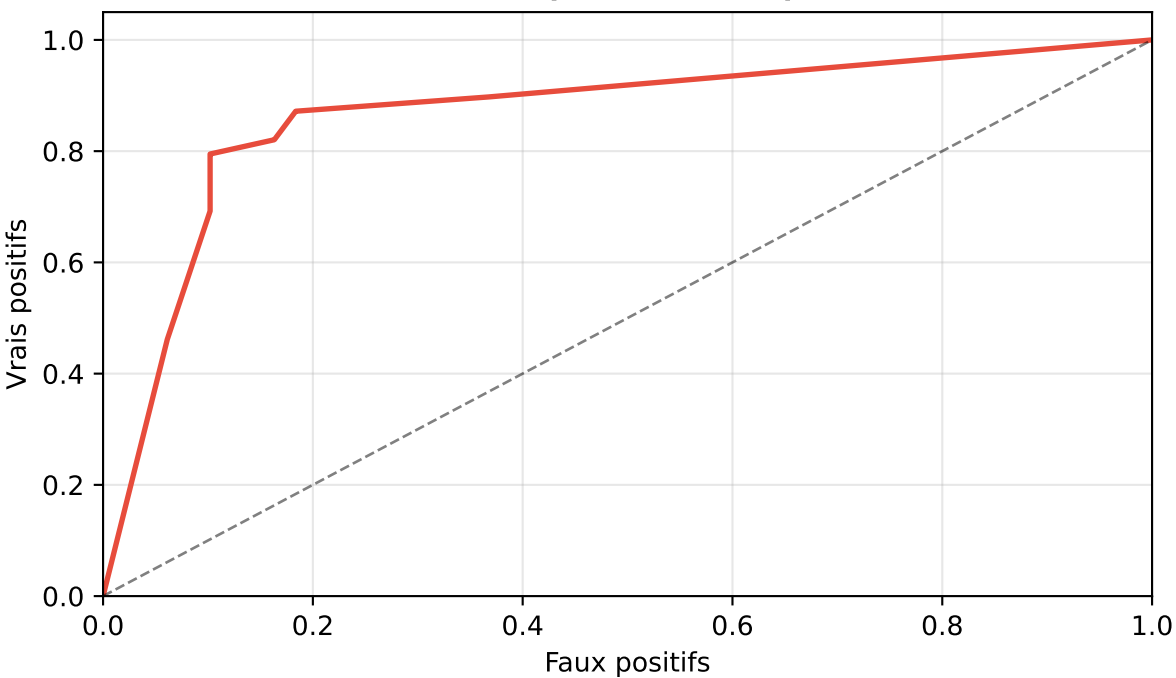
TP: 32

TN: 41

FP: 8

FN: 7

ROC (AUC = 0.867)



Détails - KNN

Métriques principales

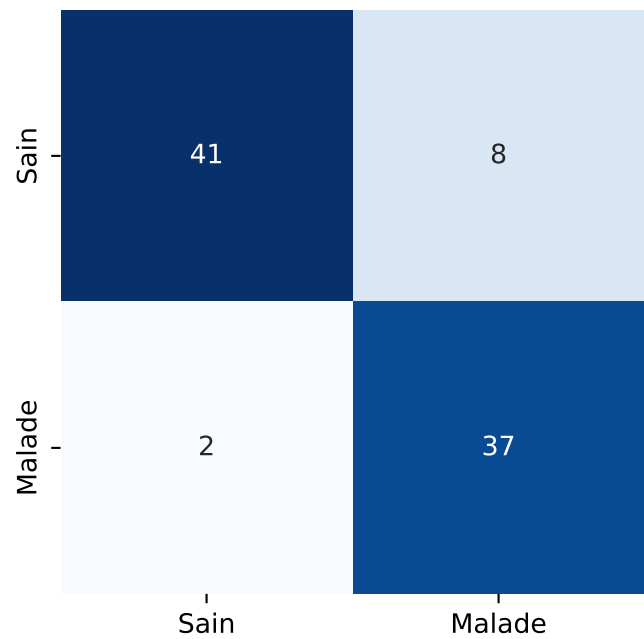
Accuracy: 88.6%

Precision: 82.2%

Recall: 94.9%

F1-Score: 88.1%

Matrice de confusion

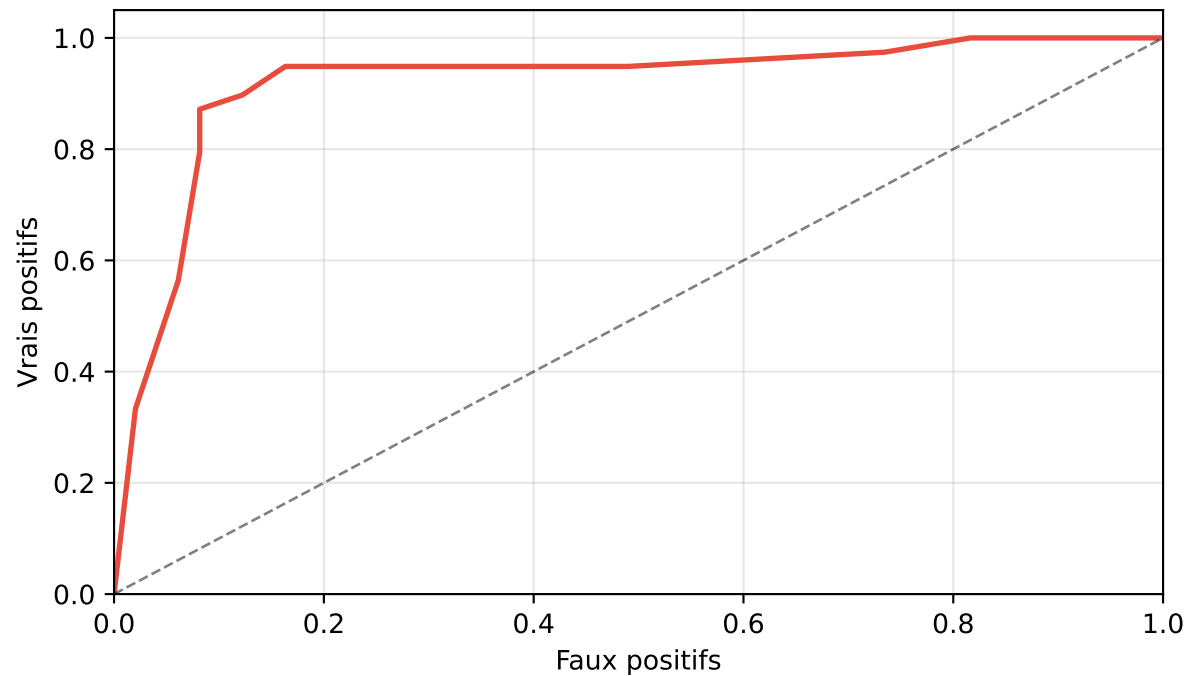


Détail matrice

TP: 37 TN: 41

FP: 8 FN: 2

ROC (AUC = 0.919)



Détails - Random Forest

Métriques principales

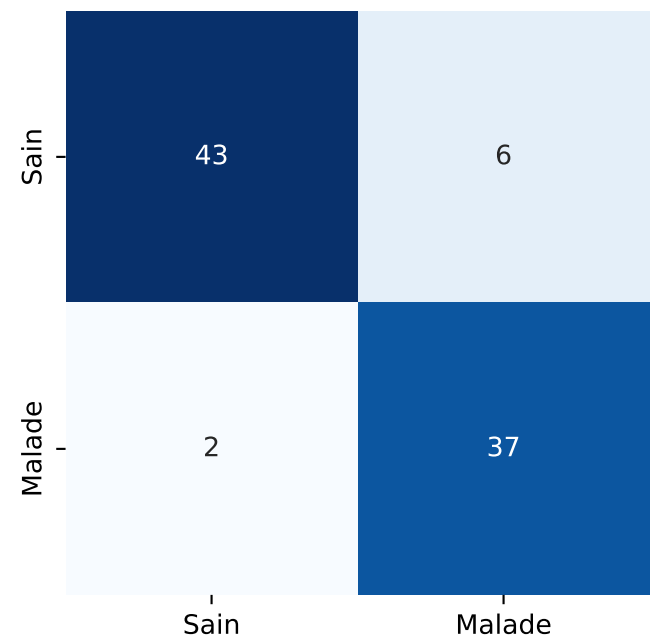
Accuracy: 90.9%

Precision: 86.0%

Recall: 94.9%

F1-Score: 90.2%

Matrice de confusion



Détail matrice

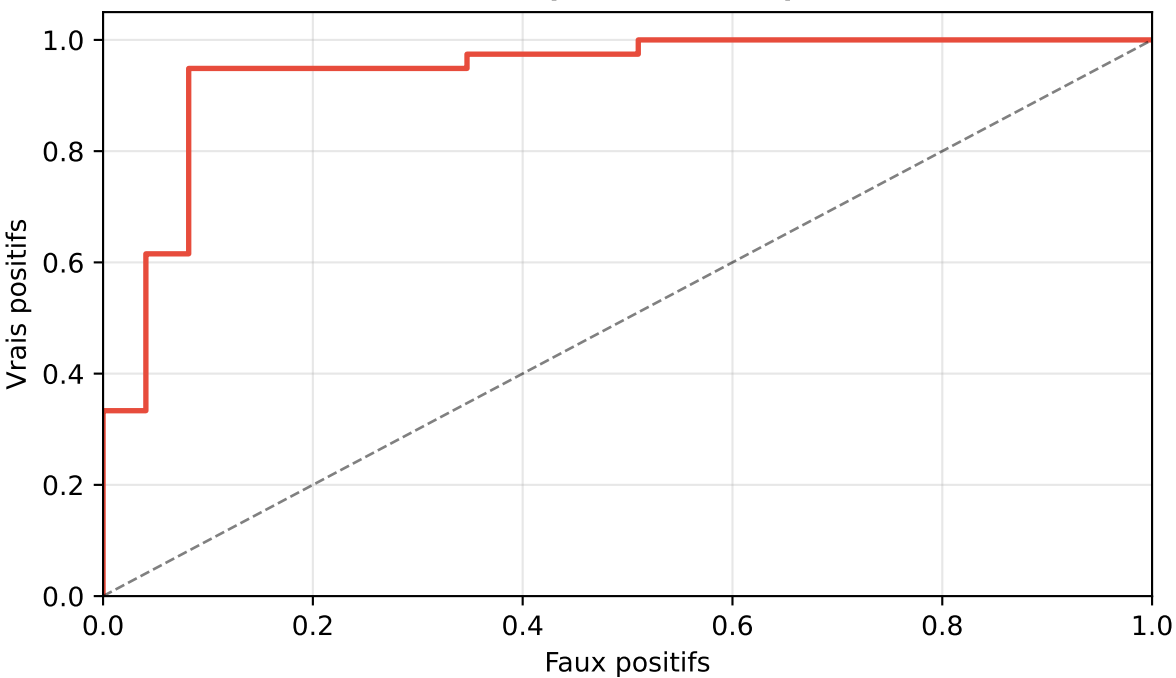
TP: 37

TN: 43

FP: 6

FN: 2

ROC (AUC = 0.939)



Détails - AdaBoost

Métriques principales

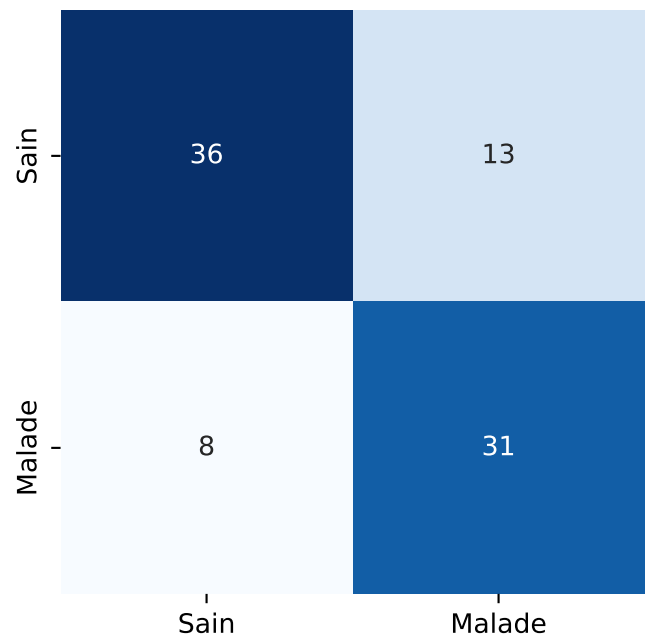
Accuracy: 76.1%

Precision: 70.5%

Recall: 79.5%

F1-Score: 74.7%

Matrice de confusion

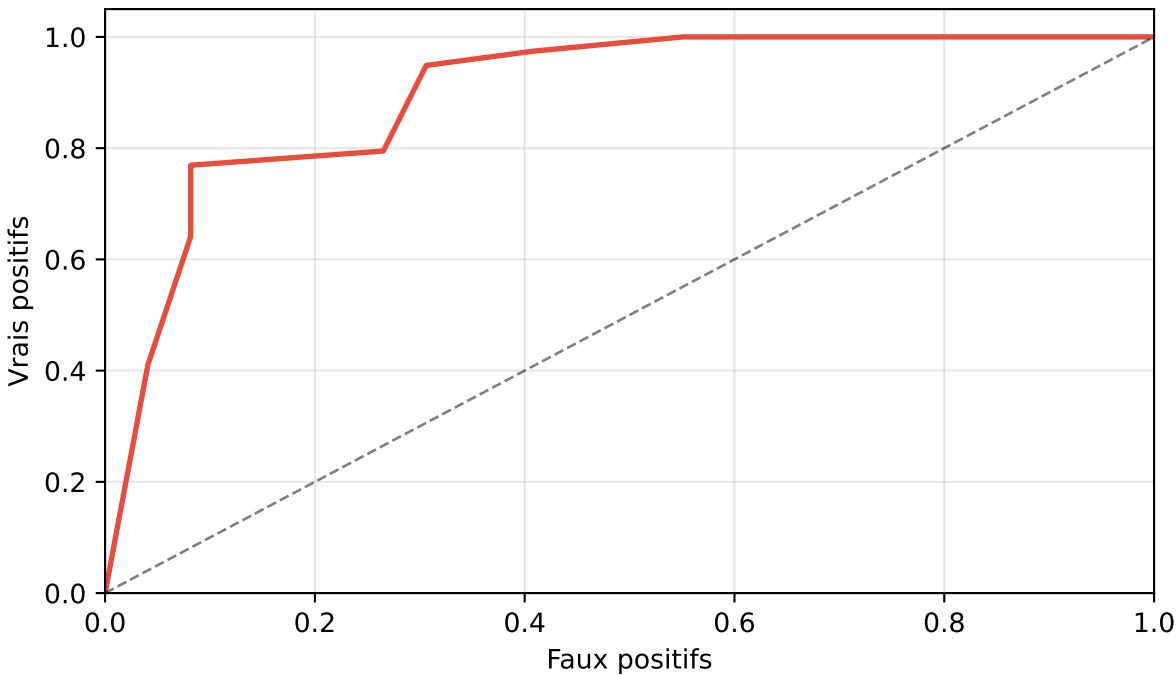


Détail matrice

TP: 31 TN: 36

FP: 13 FN: 8

ROC (AUC = 0.897)



Détails - SVM

Métriques principales

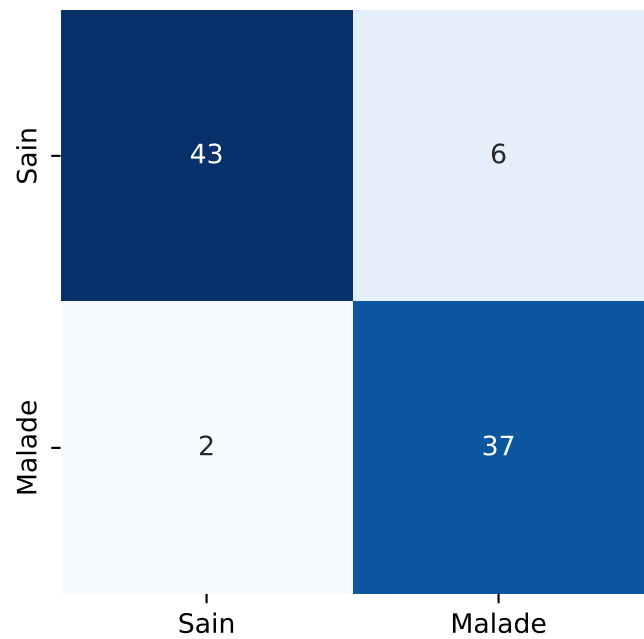
Accuracy: 90.9%

Precision: 86.0%

Recall: 94.9%

F1-Score: 90.2%

Matrice de confusion



Détail matrice

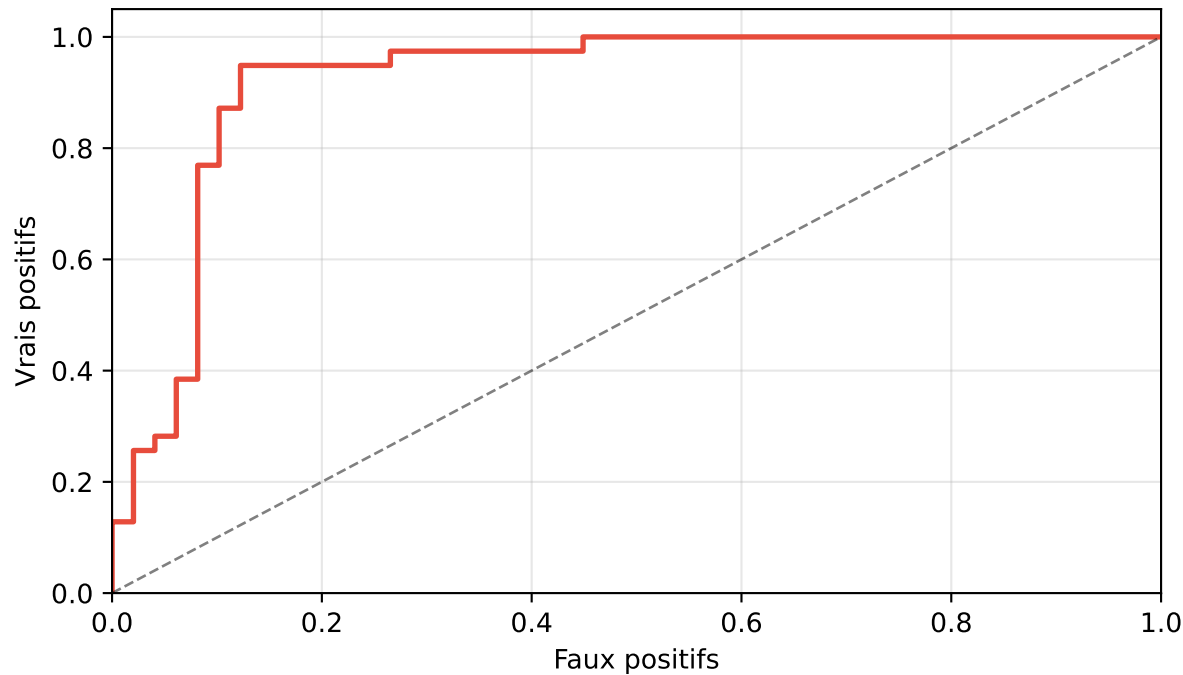
TP: 37

TN: 43

FP: 6

FN: 2

ROC (AUC = 0.920)



Détails - Voting Ensemble

Métriques principales

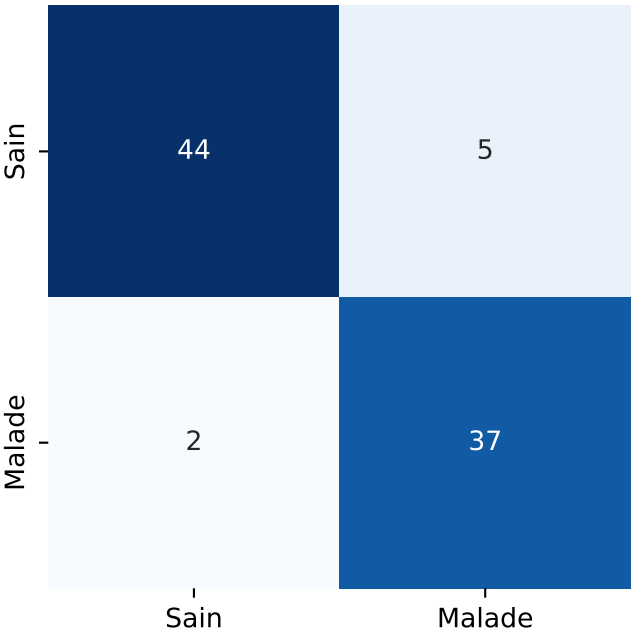
Accuracy: 92.0%

Precision: 88.1%

Recall: 94.9%

F1-Score: 91.4%

Matrice de confusion



Détail matrice

TP: 37

TN: 44

FP: 5

FN: 2

ROC (AUC = 0.934)

